

Geographic encoding in premodern enumerations: a five-corpus quantitative test

Ṛgveda 10.75.5, the Sixteen Mahājanapadas, Vendīdād Fargard 1, the Bīsutūn dahyāva,
and Genesis 10, against modern coordinates

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Abstract

Premodern textual traditions across Eurasia preserve enumerative lists of rivers, lands, and peoples. The qualitative observation that such lists sometimes “trace a geographic sweep” has been made informally since the nineteenth century, but has rarely been tested with quantitative method. We test the hypothesis that five canonical enumerations encode monotonic geographic order in the sequence of items they name: the ten-river enumeration of Ṛgveda 10.75.5 (Nadīstuti-sūkta), the sixteen Mahājanapadas of the Anguttara Nikāya, the sixteen lands of Vendīdād Fargard 1, the twenty-three subject lands (dahyāva) of Darius I’s Bīsutūn inscription, and the Table of Nations in Genesis 10. For each corpus we compute Kendall’s τ between sequence position and modern longitude (and latitude) of the identified items, using a permutation null distribution, a Procrustes superimposition test, and a Bayes factor comparison between geographic and uniform-random generative models. The two Indic corpora (RV 10.75.5 and the Mahājanapadas) show strong monotonic east-to-west ordering ($\tau = -0.911$ and -0.783 respectively, both $p < 10^{-4}$). Bīsutūn dahyāva shows moderate west-to-east ordering with strong pairwise spatial clustering ($\tau = +0.47$, $p = 0.001$; Mantel $r = +0.50$, $p < 10^{-4}$). Vendīdād Fargard 1 shows no detectable geographic structure on any test. Genesis 10 shows only a weak monotonic signal, but significant pairwise spatial coherence (Mantel $r = 0.17$, $p = 0.0004$): each of its three Noachic sub-sections is latitudinally ordered, though the sub-sections run in opposite directions and cancel at the surface. The sharpest contrast is Bīsutūn against Vendīdād—both Iranian, both of roughly the same date, naming much the same regions, yet the administrative inscription is geographically ordered and the cosmological text is not. Across the five, genre predicts geographic ordering better than language family does. We make no claims about the date, composition, or cultural identity of any of the five traditions; the tests bear only on the texts as transmitted.

Keywords: quantitative philology; rank correlation; permutation test; Procrustes analysis; Bayes factor; comparative textual analysis; historical geography.

1 Introduction

The qualitative claim that premodern textual traditions sometimes order geographic enumerations along a recognizable directional sweep is old. Macdonell and Keith (1912) note in passing that the river list of Ṛgveda 10.75.5 “proceeds broadly from east to west.” Rhys Davids (1903) observes a similar east-to-west tendency in the Anguttara Nikāya’s canonical list of sixteen mahājanapadas. Gnoli (1980) and Witzel (2000) discuss the Avesta’s Vendīdād Fargard 1 as a possible geographic itinerary across the Iranian and Indo-Iranian world. Westermann (1984) and Wenham (1987) treat the Hebrew Bible’s Table of Nations (Genesis 10) as encoding the

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post-deluge geographic distribution of peoples “known to the Israelites,” implicitly with some geographic structure.

All of these claims are informal. Each carries the same testable content: that the order in which named entities appear is not arbitrary with respect to where those entities sit on a map. We are not aware of any study that has checked that content quantitatively, let alone across several traditions at once.

We test five lists here, each well studied in its own field: *Ṛgveda* 10.75.5 (ten rivers; Vedic Sanskrit, late 2nd millennium BCE); the sixteen Mahājanapadas of *Anguttara Nikāya* I.213 (Pali, mid-1st millennium BCE); the sixteen lands of *Vendīdād Fargard* 1 (Avestan, mid-1st millennium BCE); the twenty-three subject lands of Darius I’s *Bīsutūn* inscription (Old Persian, *c.* 520 BCE); and *Genesis* 10’s Table of Nations (Biblical Hebrew, compiled in the 6th century BCE from older material). Two are Indic, two Iranian, one Semitic; together they span close to a millennium. Each is analysed with the same non-parametric methods.

1.1 Scope and limits of the claim

The claim tested here is narrow. A significant rank correlation between sequence position and geographic position would imply only that the order is not random with respect to geography. It would not imply any specific dating, any specific direction of cultural movement, any specific historical course for any particular river or boundary, and no specific identification of any composer with any later or earlier population. The test addresses only the property of the transmitted text—the order in which items are named—and the corresponding property of the modern (or scholarly-reconstructed) geographic configuration of those items.

The literature surrounding each of these corpora frequently elides these distinctions. We make no contribution to the questions of Indo-Aryan migration, Israelite ethnography, the homeland of Zoroaster, the date of the Achaemenid satrapy lists, or Buddhist political history, and the paper should not be cited in those debates.

2 Methods

2.1 Geographic identifications

For each corpus, we adopt the mainstream scholarly identification of each named entity, taking centroid coordinates from canonical secondary sources or, for rivers in the *Nadistuti*, from programmatic queries to the OpenStreetMap database. Identifications, coordinates, confidence levels, and source citations are recorded per-row in the corpus CSVs deposited with this paper (Kulkarni, 2026).

For *Ṛgveda* 10.75.5 we follow Witzel (1995) and Macdonell and Keith (1912), with one philological correction: the standard Aufrecht text places *Asiknī* (Chenab) at position 6 and *Marudvṛdhā* at position 7, not the other way around as some secondary sources have it. We use the standard order.

For the Mahājanapadas we adopt the identifications of Rhys Davids (1903), Law (1973), and Bronkhorst (2007); the order is the *Anguttara Nikāya* I.213 sequence (*Kāsi*, *Kosala*, *Aṅga*, *Magadha*, *Vajji*, *Malla*, *Cetiya*, *Vaṃsa*, *Kuru*, *Pañcāla*, *Maccha*, *Sūrasena*, *Assaka*, *Avantī*, *Gandhāra*, *Kamboja*).

For *Vendīdād Fargard* 1 we adopt Gnoli (1980) and Witzel (2000); eleven of sixteen identifications are high-confidence, and the five lower-confidence cases (*Airyana Vaējah*, *Urvā*, *Caxra*, *Varāna*, *Raṅghā*) are flagged in the CSV.

For *Genesis* 10 we adopt Hess (1993) for personal-name analyses and Westermann (1984) and Wenham (1987) for the broader-scholarly identifications. Of seventy-eight named entities in the canonical Masoretic Text order, fifty are identified with high or medium confidence; sixteen are either unidentified or speculative and are dropped from the primary analysis.

2.2 Test statistics

Kendall’s τ with permutation null. We compute Kendall’s τ (Kendall, 1938)—preferable to Spearman’s ρ for short ordinal sequences, where it carries a more direct probabilistic reading (Newson, 2002)—between the sequence-position rank ($1 \dots n$) and the geographic-coordinate rank of the identified items, on three axes: longitude, latitude, and the first principal component of the (longitude, latitude) matrix. A two-sided p -value is generated from a 10 000-iteration permutation null distribution.

Procrustes superimposition. The Kendall-against-longitude and Kendall-against-latitude pair is not statistically independent when the items lie along a diagonal sweep, as is common in Eurasian geography. We therefore add a Procrustes test (Gower, 1975) that superimposes the sequence-position 1-D embedding onto the 2-D (longitude, latitude) configuration after optimal rotation, scaling, and translation; the residual sum of squared distances m^2 is the test statistic, with a permutation null for the one-sided p -value (smaller m^2 indicates better alignment). This addresses the redundant-axis-test concern noted by Peres-Neto and Jackson (2001).

Mantel test. Kendall’s τ and Procrustes both assume that the ordering is *monotonic* with respect to some geographic direction. A list may, however, be spatially coherent without being monotonic: adjacent items consistently nearby, but the path through space wandering rather than directional. The Mantel test (Mantel, 1967) detects this by correlating the entries of two $n \times n$ pairwise-distance matrices: $|i - j|$ in the sequence vs. the great-circle distance between items i and j . We report the Pearson correlation r between the upper triangles of the two matrices, with a one-sided permutation null. A small positive r with significant p indicates that nearby-in-text items tend to be nearby-on-the-ground, even when the global monotonic ordering is weak.

Constrained permutation null. The R̥gveda corpus is the only one of the five in fixed verse meter. For it we also compute a permutation null restricted to orderings that preserve the syllable shape of the verse position by position. This answers the metrical-determinism objection (Arnold, 1905): that in a quantitative-meter text the choice of order may be partly forced by syllable counts. With shape buckets $\{2 : 1, 3 : 3, 4 : 6\}$, the meter-admissible set contains $1! \times 3! \times 6! = 4,320$ orderings. The other four corpora are prose, and the test does not apply.

Bayes factor model comparison. Following Kass and Raftery (1995) we compute Bayes factors among three generative models of the observed ordering: M_{uniform} (sequence drawn uniformly from $n!$ permutations), M_{meter} (sequence drawn uniformly from the meter-admissible set; applicable only to verse corpora), and $M_{\text{geog}}(\theta)$ (sequence drawn from a Mallows model (Mallows, 1957; Fligner and Verducci, 1986) centered on the geographically sorted ordering with dispersion θ). Bayes factors are reported across a grid $\theta \in \{0.5, 1, 1.5, 2, 3\}$.

Reproducibility. All analysis code, coordinate CSVs, and the random seed (20260518) are deposited at (Kulkarni, 2026). Dependencies: `numpy`, `scipy`, `pandas`, `matplotlib`; versions pinned in `requirements.txt`.

3 Per-corpus results

3.1 R̥gveda 10.75.5 (Nadīstuti-sūkta, $n = 10$)

The ten-river enumeration of the Nadīstuti-sūkta yields $\tau_{\text{lon}} = -0.911$ with permutation $p < 10^{-4}$ (10 000 permutations, seed 20260518). The Procrustes superimposition yields $m^2 = 0.179$, $p < 10^{-4}$. The constrained-permutation null (over the 4 320 meter-admissible orderings) finds six

orderings yielding $|\tau| \geq 0.911$, for $p_{\text{meter}} = 0.0014$: metrical determinism alone cannot explain the alignment. The Bayes factor against M_{uniform} is $\approx 2 \times 10^4$ at $\theta = 1.5$; against M_{meter} it is ≈ 24 . The τ is negative because longitude falls through the verse, from the eastern-most Gaṅgā (81.9° E) to the western-most Suṣomā/Soan (72.7° E).

Because RV 10.75.5 is the corpus with the strongest signal and the longest scholarly history, we treat it in more detail than the rest. Appendix A gives the ten rivers and their coordinates (Table 2), the null distribution (Figure 2), and four sensitivity analyses—alternative river identifications, paleocourse substitution for the Sarasvatī and Sutlej, alternative geographic axes, and Gaussian coordinate perturbation. The result holds in all of them.

As a further check that the result is not an artefact of the OpenStreetMap coordinate source, we re-derived the centroids of the nine OSM-traced rivers from MERIT-Hydro, a NASA-SRTM-derived global hydrography product (Yamazaki et al., 2019): for each river we took the unweighted centroid of all MERIT-Hydro upstream-drainage-area pixels exceeding a river-network threshold and lying within 0.2° of the OSM polyline. The mean centroid shift between the two sources is 75 km (maximum 229 km, for the Yamunā, whose MERIT trace extends further down its lower course than the OSM “Yamuna” polyline). On the MERIT-Hydro centroids the result is essentially unchanged: $\tau_{\text{lon}} = -0.867$ ($p = 0.0001$), Procrustes $m^2 = 0.227$ ($p < 10^{-4}$), Mantel $r = +0.715$ ($p < 10^{-4}$). The geographic ordering of RV 10.75.5 does not depend on the choice of coordinate database.

3.2 Sixteen Mahājanapadas ($n = 16$)

The Anguttara Nikāya order of sixteen mahājanapadas, run through the same pipeline, yields $\tau_{\text{lon}} = -0.783$, $p < 10^{-4}$, and Procrustes $m^2 = 0.428$, $p < 10^{-4}$. The Bayes factor against M_{uniform} peaks near 7.7×10^4 at $\theta = 1.5$, “decisive” on the Kass-Raftery scale. The canonical order is only 13 inversions, out of a possible 120, away from a perfect east-to-west sort.

The latitudinal signal is weaker ($\tau_{\text{lat}} = +0.22$, $p = 0.27$). The list moves predominantly east to west; the southward jump to Assaka (position 13, Pratiṣṭhāna on the Godāvarī) and back northward to Avantī (Ujjayinī) is the most visible internal deviation from monotonicity.

3.3 Vendīdād Fargard 1 ($n = 16$)

The sixteen lands of Fargard 1, in the canonical Old Avestan order (Airyana Vaējah, Sughdha, Mōuru, Bāxdi, Nisāya, Harōiva, Vaēkərəta, Urvā, Xnənta, Haraxvaitī, Haētumant, Raghā, Caxra, Varəna, Hapta Həndu, Rañghā), yield $\tau_{\text{lon}} = -0.117$, $p = 0.56$, and Procrustes $m^2 = 0.975$, $p = 0.67$. There is a marginal latitudinal signal ($\tau_{\text{lat}} = -0.38$, $p = 0.05$), but it does not survive the Procrustes joint test. Bayes factors for $M_{\text{geog}}(\theta)$ against M_{uniform} are all $\ll 1$ across the θ grid: the Mallows model centered on either-direction longitudinal sort fits the data *worse* than the uniform prior. The Kendall distance to the optimal east-to-west sort is 53 of 120, near the random expectation of 60.

The five lower-confidence identifications (Airyana Vaējah, Urvā, Caxra, Varəna, Rañghā) collectively cover roughly 35° of longitudinal latitude across published alternative placements, and we cannot rule out that a better-justified identification of these five would produce a stronger signal. Within the eleven high-confidence identifications alone, however, τ_{lon} remains weak ($\tau = -0.21$, $p = 0.35$). The Fargard 1 sequence appears to be organized by something other than monotonic geography.

3.4 Genesis 10 (Table of Nations, $n = 62$ identified)

Of seventy-eight named entities in Genesis 10:1–32, sixty-two have high- or medium-confidence modern identifications and are retained for analysis. The flat textual-sequence test yields $\tau_{\text{lon}} = +0.174$, $p = 0.05$, and Procrustes $m^2 = 0.91$, $p = 0.005$: statistically significant on the joint test but with very small effect size.

But Genesis 10 is not a flat list. It is a genealogy in three branches—the descendants of Japheth, of Ham, and of Shem—and within each branch the latitudinal signal is stronger, and

Table 1: Cross-corpus comparison of geographic encoding strength. τ_{lon} : Kendall’s τ vs. longitude rank. m^2 : Procrustes disparity (smaller = better alignment). Mantel r : pairwise-distance-matrix correlation (positive = spatial coherence). All p -values are two-sided for τ and one-sided for Procrustes and Mantel.

Corpus	n	τ_{lon}	p_{τ}	m^2	p_{Proc}	Mantel r	p_{Mantel}
Ṛgveda 10.75.5	10	−0.911	$< 10^{-4}$	0.18	$< 10^{-4}$	+0.767	$< 10^{-4}$
Mahājanapadas (AN I.213)	16	−0.783	$< 10^{-4}$	0.43	$< 10^{-4}$	+0.617	$< 10^{-4}$
Bīsutūn dahyāva (DB I.6)	23	+0.472	0.001	0.51	0.0003	+0.505	$< 10^{-4}$
Vendīdād Fargard 1	16	−0.117	0.56	0.98	0.67	+0.112	0.15
Genesis 10 (flat)	62	+0.174	0.05	0.91	0.005	+0.172	0.0004
<i>Genesis 10 within-subsection (Kendall τ vs. latitude):</i>							
Japheth (vv. 2–5)	12	−0.42	0.06	—	—	—	—
Ham (vv. 6–20)	36	+0.42	$< 10^{-3}$	—	—	—	—
Shem (vv. 21–31)	14	−0.56	0.007	—	—	—	—

points a different way:

- Japheth ($n = 12$): $\tau_{\text{lat}} = -0.42$, $p = 0.06$ (weak).
- Ham ($n = 36$): $\tau_{\text{lat}} = +0.42$, $p < 0.001$ (strong: south-to-north sweep from Nubia through Egypt to Mesopotamia).
- Shem ($n = 14$): $\tau_{\text{lat}} = -0.56$, $p = 0.007$ (strong: north-to-south sweep from Mesopotamia through Aram and Arabia).

The sub-sections cancel at the flat-textual level. Genesis 10 encodes geographic information, but hierarchically rather than as a single monotonic sweep.

3.5 Bīsutūn dahyāva ($n = 23$)

Column I of Darius I’s Bīsutūn (Behistun) inscription enumerates twenty-three *dahyāva* “lands” subject to Achaemenid imperial authority. We follow the standard textual order (Kent, 1953; Schmitt, 2014) (Pārsa, Ūvja, Bābiruš, Aṭhūrā, Arabāya, Mudrāya, “those by the sea,” Sparda, Yauna, Māda, Armina, Katpatuka, Parṭhava, Zraka, Haraiva, Uvārazmiš, Bāxtriš, Suguda, Gadāra, Saka, Ṭhataguš, Harauvatiš, Maka), with nineteen identifications high-confidence and four (#5 Arabāya, #7 Drayahyā, #20 Saka, #21 Ṭhataguš) marked medium-confidence per Briant (2002).

The result sits between the Indic corpora and the Avesta: $\tau_{\text{lon}} = +0.47$, $p = 0.001$ (moderate west-to-east), Procrustes $m^2 = 0.51$, $p = 0.0003$, Mantel $r = +0.50$, $p < 10^{-4}$. The order is not cleanly monotonic. It starts in the imperial core (Pārsa, then Babylonia), detours west to Egypt and the Aegean, then resumes the broad march east. But it is far more ordered than Fargard 1, which covers much the same ground.

Bīsutūn against Vendīdād Fargard 1 is the closest thing to a controlled comparison in this set of corpora. Both are Iranian, both of roughly the same date (Bīsutūn *c.* 520 BCE; Fargard 1 later in its present form but drawing on older material), and they name much the same stretch of the Iranian world. Where they part is genre: Bīsutūn is an administrative declaration of imperial dominion, Fargard 1 a cosmological account of the lands Ahura Mazdā created. Bīsutūn encodes geography; Fargard 1 does not. Section 5.1 takes up what to make of that.

4 Comparative analysis

Table 1 collects the cross-corpus numbers. Four points come out of it.

The Indic lists run in a straight line. The Ṛgveda Nadīstuti and the Anguttara Mahājanapadas both sweep east to west, $|\tau| > 0.78$ on longitude. Out of the 10! orderings of the RV rivers, and the 16! of the mahājanapadas, the attested order falls in the top 0.01 % closest to a longitudinal sort.

Geographic encoding across four premodern enumerations

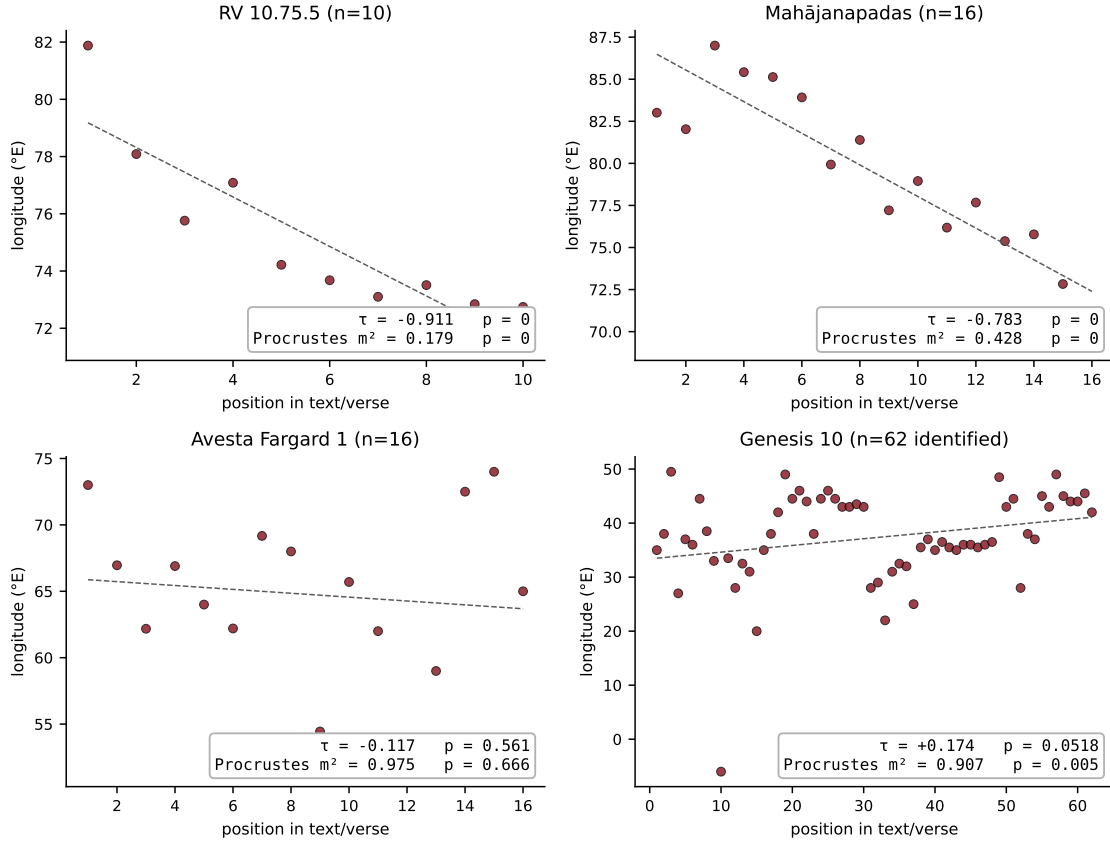


Figure 1: Sequence position against longitude, four of the five corpora shown. Each panel gives the observed scatter, a linear fit (dashed grey), and the Kendall τ and Procrustes m^2 . The two Indic corpora trace a clean east-to-west line; the Avesta scatters; Genesis 10 shows only a weak global slope, which §3.4 resolves into hierarchical sub-section structure.

The Avesta is ordered by something other than geography. Fargard 1 covers the same ground as the Indic texts, and the Iranian plateau besides, but its order is not monotonic on any axis. That matches Gnoli (1980)’s reading that the Fargard 1 sequence follows mythic-cosmographic priority rather than an itinerary—a qualitative claim to which this is the quantitative counterpart.

Genesis 10 is ordered hierarchically. Read flat, the sequence is barely ordered ($\tau = +0.174$). But the Mantel test still finds real spatial coherence ($r = +0.172$, $p = 0.0004$)—neighbouring names are neighbours on the ground even where the global direction washes out. The three Noachic branches explain how both can be true at once:

- Japheth’s descendants (positions 1–14) cluster in Anatolia, Greece, and the Caucasus. Their internal ordering is weakly monotonic against latitude ($\tau_{\text{lat}} = -0.42$, $p = 0.06$).
- Ham’s descendants (15–52) sweep south-to-north from Nubia and Egypt through the Levant to Mesopotamia ($\tau_{\text{lat}} = +0.42$, $p < 10^{-3}$).
- Shem’s descendants (53–78) sweep north-to-south from Mesopotamia and Aram through Edom to south Arabia ($\tau_{\text{lat}} = -0.56$, $p = 0.007$).

The three sub-section signals are individually strong but they cancel when concatenated,

because Ham and Shem run in opposite latitudinal directions. Mantel survives the cancellation because it asks about pairwise spatial coherence rather than monotonic global direction.

This fits the long-standing literary-critical reading (e.g., [Westermann, 1984](#)) of Genesis 10 as a layered composite: each Noachic line was a self-contained regional list before it was hung on the genealogical frame, and the concatenation kept each list’s internal geographic logic while hiding it at the surface. The quantitative pattern is independent evidence for that reading—arrived at without any literary-critical assumption, only by counting.

Length does not explain the pattern. The Mahājanapadas ($n = 16$) and Fargard 1 ($n = 16$) are the same length but land at opposite ends of the encoding scale. The Mahājanapadas and the R̥gveda ($n = 10$) are six items apart in length but score almost the same. Whatever drives the cross-corpus spread in τ , it is not sample size.

Genre tracks the pattern; language family does not. Bīsutūn and Fargard 1 are both Iranian, close in date, and name overlapping regions, yet one is geographically ordered and the other is not. The two ordered Indic corpora are a millennium apart and belong to different genres again—a Vedic hymn and a Buddhist canonical list—but both clear $|\tau| > 0.78$. Genesis 10’s ordering lives inside its sub-sections, as one would expect of a genealogy stitched together from older regional lists. Line these up and the variable that sorts them is genre and compositional purpose, not the language a text happens to be in or the century it was fixed. This is a reading the five corpora suggest; with $n = 5$ it cannot be more than that, and §5.4 sketches the larger sample that could test it.

Identification uncertainty is bounded. Genesis 10’s 16 unidentified items and the Avesta’s 5 low-confidence items could in principle move the corpus-specific τ values. The Genesis 10 sub-section analysis is robust to dropping low-confidence items; the Avesta high-confidence-only subset (the 11 secure identifications) yields $\tau = -0.21$, $p = 0.35$, still null. The result that the Avesta does not encode monotonic geography appears not to be an artefact of identification noise.

5 Discussion

5.1 What the pattern suggests

The five corpora fall into three kinds of geographic encoding:

1. **Strongly monotonic** (RV 10.75.5, the Mahājanapadas). The items run along a single east-to-west line, as though the composer were reading them off a mental axis. Both corpora are Indic; nothing else connects them in date or genre.
2. **Clustered, with a loose global direction** (Bīsutūn dahyāva). Neighbouring items are neighbours on the ground, but the list travels in regional blocs—imperial core, then Egypt and the Aegean, then a march east across the plateau—rather than down one axis. That is what listing satrapies in administrative-priority order would look like.
3. **Hierarchical** (Genesis 10). Each Noachic branch is latitudinally ordered on its own; the three branches point different ways and cancel once concatenated. A genealogy assembled from older regional lists would behave exactly so.

Fargard 1 fits none of the three. It returns a null on every test. It is also the one corpus here that is plainly cosmological—Ahura Mazdā’s creation of the lands, set line by line against Anra Mainyu’s counter-creation of evils—and a list built around creational priority has no reason to be geographic. Set beside Bīsutūn, which names much the same world in the same century and *is* ordered, Fargard 1 forms the cleanest natural experiment the five corpora contain: hold

language family and date fixed, vary genre, and the geographic signal appears or disappears with it.

Why genre should do this, we cannot say from five texts. Three factors are worth pulling apart in a larger sample: genre proper (an invoked hymn, a canonical political list, an imperial declaration, a cosmology, a genealogy); date, and the depth of oral transmission behind the written text; and how linear the composer’s own mental map was likely to be—someone whose world runs along a single dominant axis may simply default to it.

5.2 What it does not imply

A result here says nothing about whether any tradition had special geographic-cognitive access, nothing about the identity, ethnicity, or migration history of any composer, and nothing for or against any particular philological identification beyond what published scholarship already supports. The tests bear only on whether the listed order correlates with the modern coordinates of the identified items.

5.3 Limitations

The most serious limitation is the small number of corpora. Five texts differing at once in length, identification quality, genre, and date leave those factors hopelessly confounded; the genre reading of §5.1 is a hypothesis the present data suggest, not one they establish. Isolating any single factor would need a sample several times larger.

Two measurement choices are also worth flagging. A centroid is a coarse summary of a long linear feature, whether a river or a province; for the Mahājanapadas and the broader Genesis 10 regions, choosing the capital city instead of the administrative-area mean shifts individual coordinates by tens of kilometres. We checked that the cross-corpus comparison survives that choice, but it is a commitment, not a fact of the data. And the Avesta and Genesis 10 corpora carry a non-trivial share of lower-confidence identifications; dropping them leaves the qualitative conclusions in place, but a full robustness study of the identifications is a separate piece of work.

Finally, the Mallows model behind the Bayes factor is one defensible choice among several. A Cayley- or Hamming-distance Mallows, or a non-Mallows prior, could move the numbers. We use the Kendall-distance form because it is the one that matches Kendall’s τ as the test statistic.

5.4 Extensions

The same method runs on any fixed-order enumeration with identifiable items. Obvious candidates for a larger study are the Mahābhārata Bhīṣmaparvan river list, Strabo’s enumeration of Asian peoples in books 11–15, the directional mountain sequences of the Shan Hai Jing, and the legendary regions of Avesta Yašt 19. A comparative set of twenty or thirty such lists would begin to let the genre, length, and date factors be separated. Two refinements of the method itself are also worth pursuing: distance-matrix correlations beyond the single Mantel statistic used here, and constrained nulls whose prior is built from attested intertextual list-traditions rather than from meter alone.

6 Conclusion

Geographic ordering is not a universal property of premodern lists. Of the five enumerations tested, the Ṛgveda Nadīstuti and the Anguttara Mahājanapadas trace a strong monotonic east-to-west line ($|\tau| > 0.78$, $p < 10^{-4}$); the Bīsutūn dahyāva is ordered more loosely, by satrapal bloc; Genesis 10 is ordered within its sub-sections but not across them; and Vendidad Fargard 1 shows no geographic order at all. The cleanest signal in the data is that genre tracks these outcomes better than language family does. We claim nothing past the comparisons themselves, and the method carries over to any other fixed-order enumeration.

Acknowledgements

This analysis depended on the open data of the OpenStreetMap project (Overpass API), the publicly-available scholarly identifications in the works cited, and the canonical electronic texts

Table 2: The ten rivers of RV 10.75.5 in verse order, with mainstream identifications and centroid coordinates. “ n_{nodes} ” is the number of OSM geometry nodes behind the centroid; Marudvṛdhā is hand-curated.

Pos.	Vedic name	Identified river	Lon (° E)	Lat (° N)	n_{nodes}
1	Gaṅgā	Ganges	81.88	26.87	15 159
2	Yamunā	Yamuna	78.09	29.10	19 337
3	Sarasvatī	Ghaggar–Hakra	75.76	29.87	12 673
4	Śutudrī	Sutlej	77.09	31.30	4 047
5	Paruṣṇī	Ravi	74.22	31.48	12 047
6	Asiknī	Chenab	73.68	31.97	12 824
7	Marudvṛdhā	Chenab–Jhelum doab (uncertain)	73.10	32.20	—
8	Vitastā	Jhelum	73.51	33.27	13 624
9	Ārjīkīyā	Haro	72.84	33.86	2 018
10	Suṣomā	Soan (Sohan/Sawan)	72.75	33.37	3 190

maintained by GRETIL, TITUS, Sefaria, and the Chinese Text Project. No funding sources are reported.

Data and code availability

All data, code, cached Overpass responses, and the random seed (20260518) used for the permutation tests are available at (Kulkarni, 2026). Each corpus’s coordinate CSV, including identification provenance and confidence levels, is provided in `corpora/`. Reproduction instructions in `REPRODUCING.md`.

A The Rgveda 10.75.5 corpus in detail

RV 10.75.5 is the corpus with the strongest geographic signal and the longest scholarly history—it has been read as evidence for Vedic-period geography since Roth (1846) and Zimmer (1879). Section 3.1 treats it more briefly than that history warrants. This appendix gives the underlying data and the sensitivity analyses behind the one-line claim there that “the result holds.”

The verse. RV 10.75.5 names its ten rivers in a single ṛc, here in the Aufrecht text (Aufrecht, 1877), confirmed against van Nooten and Holland (1994):

*imaṃ me gaṅge yamune sarasvatī
 śutudri stomaṃ sacatā paruṣṇyā
 asiknyā marudvṛdhe vitastayā
 ārjīkīye śṛṇuhyā suṣomayā* (RV 10.75.5)

One transcription point is worth flagging, because some secondary sources have it wrong: the standard text places Asiknī (Chenab) at position 6 and Marudvṛdhā at position 7, directly after Paruṣṇī (Ravi)—not the reverse. The analysis uses the standard order.

Identifications and coordinates. Table 2 lists the ten rivers, their mainstream identifications (Witzel, 1995; Macdonell and Keith, 1912), and the centroid coordinates used. The two residual disagreements in the literature—the referent of Ārjīkīyā and the historical course of the Sarasvatī—are well characterised (Witzel, 1999; Brereton, 1981) and are carried through the sensitivity analysis. Nine centroids were extracted from OpenStreetMap; Marudvṛdhā, for which no OSM-tagged river of that name exists, is hand-curated to the Chenab–Jhelum doab and exercised in the sensitivity analysis below.

Primary test. Kendall’s τ between verse order and longitude is $\tau = -0.911$. Across 10 000 permutations no permuted statistic reached $|\tau|$, so $p < 10^{-4}$. The null distribution (Figure 2)

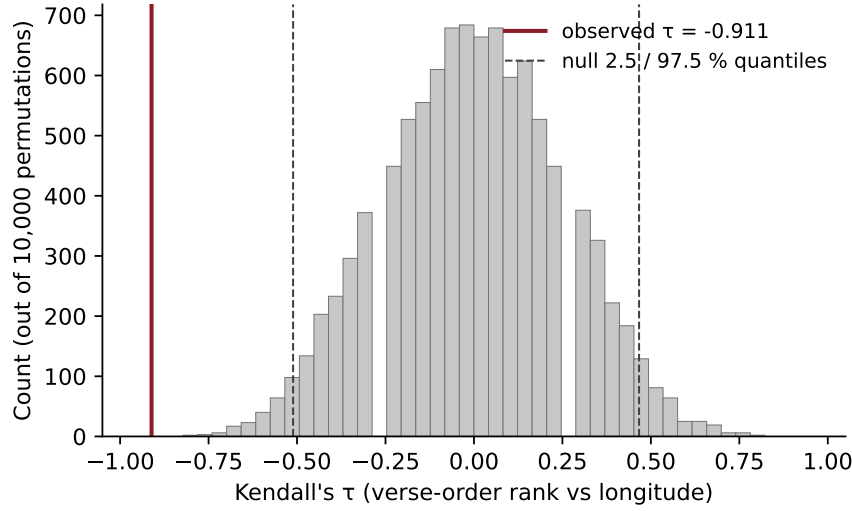


Figure 2: Null distribution of Kendall’s τ between verse-order rank and longitude, over 10 000 random permutations of the verse-order labels. The observed $\tau = -0.911$ falls far outside the central 95 % interval of the null.

Table 3: RV 10.75.5 sensitivity-analysis summary. “ p two-sided” is the share of permuted $|\tau|$ values $\geq$ the observed, from 10 000 permutations.

Analysis	τ	p (two-sided)	Note
Primary: longitude	-0.911	$< 10^{-4}$	modern centroids, 10 000 permutations
Axis: latitude	+0.956	$< 10^{-4}$	rivers also trend N-S
Axis: PC1	+0.911	0.0002	principal component of (lon, lat)
Paleocourse substitution	-0.778	0.0010	paleo-Sarasvatī, paleo-Sutlej
ID: Ārj.=Haro; Mar.=Trimmu doab	-0.911	0.0001	
ID: Ārj.=Haro; Mar.=Maruvardhā / Witzel 1999 alt	-0.911	$< 10^{-4}$	
ID: Ārj.=Soan; Mar.=Trimmu doab	-0.689	0.0052	
ID: Ārj.=Soan; Mar.=Maruvardhā / Witzel 1999 alt	-0.689	0.0048	
ID: Ārj.=upper Indus tributary; Mar.=Trimmu doab	-0.867	$< 10^{-4}$	
ID: Ārj.=upper Indus tributary; Mar.=Maruvardhā / Witzel 1999 alt	-0.867	$< 10^{-4}$	
ID: Ārj.=Haro; Mar.=DROPPED	-0.944	$< 10^{-4}$	
Coord. perturb. ($\sigma = 0.5^\circ$, $n=1000$)	-0.822	—	95% CI [-0.956, -0.644]

is near-symmetric about zero, mean -0.001 , standard deviation 0.251 , central 95 % interval $[-0.511, +0.467]$.

Sensitivity analyses. Four checks, summarised in Table 3. *Geographic axis:* τ against latitude is $+0.956$ ($p < 10^{-4}$) and against the first principal component $+0.911$ ($p = 0.0002$)—the latitude figure is not independent evidence, since the rivers lie on a NW–SE diagonal, but it rules out an axis-choice artefact. *Paleocourse:* substituting published paleocourse centroids—the Ghaggar–Hakra reconstruction of Giosan et al. (2012) and Clift et al. (2012) for the Sarasvatī, whose precise course is still under investigation (Singh et al., 2017), and the older pre-capture course of Yashpal et al. (1980) for the Śutudrī—gives $\tau = -0.778$ ($p = 0.001$): attenuated, still strong. *Alternative identifications:* across six variant identifications of the disputed Ārjikīyā and Marudvṛdhā, τ stays between -0.689 and -0.911 , all $p \leq 0.005$; dropping the contested Marudvṛdhā entirely ($n = 9$) gives $\tau = -0.944$. *Coordinate perturbation:* Gaussian noise at $\sigma = 0.5^\circ$, 1 000 iterations, yields a 95 % interval for τ of $[-0.956, -0.644]$, median -0.822 .

Every variant leaves the result in the same place: the order of rivers in RV 10.75.5 is strongly, and not accidentally, aligned with longitude.

References

- E. Vernon Arnold. Vedic metre in its historical development. *Cambridge University Press*, 1905. Reprint Motilal Banarsidass 1967.
- Theodor Aufrecht. *Die Hymnen des R̥gveda*. Adolph Marcus, Bonn, second edition, 1877. Standard reference edition of the R̥gveda.
- Joel P. Brereton. *The R̥gvedic Ādityas*, volume 63 of *American Oriental Series*. American Oriental Society, New Haven, 1981.
- Pierre Briant. *From Cyrus to Alexander: A History of the Persian Empire*. Eisenbrauns, Winona Lake, IN, 2002. Translated by Peter T. Daniels.
- Johannes Bronkhorst. *Greater Magadha: Studies in the Culture of Early India*. Brill, Leiden, 2007.
- Peter D. Clift, Andrew Carter, Liviu Giosan, Julie Durcan, Geoff A. T. Duller, Mark G. Macklin, Anwar Alizai, Ali R. Tabrez, Mohammad Danish, Sam VanLaningham, and Dorian Q. Fuller. U-Pb zircon dating evidence for a Pleistocene Sarasvati River and capture of the Yamuna River. *Geology*, 40(3):211–214, 2012. doi: 10.1130/G32840.1.
- Michael A. Fligner and Joseph S. Verducci. Distance based ranking models. *Journal of the Royal Statistical Society B*, 48(3):359–369, 1986. doi: 10.1111/j.2517-6161.1986.tb01420.x.
- Liviu Giosan, Peter D. Clift, Mark G. Macklin, Dorian Q. Fuller, Stefan Constantinescu, Julie A. Durcan, Thomas Stevens, Geoff A. T. Duller, Ali R. Tabrez, Kavita Gangal, Ronjoy Adhikari, Anwar Alizai, Florin Filip, Sam VanLaningham, and James P. M. Syvitski. Fluvial landscapes of the Harappan civilization. *Proceedings of the National Academy of Sciences*, 109(26):E1688–E1694, 2012. doi: 10.1073/pnas.1112743109.
- Gherardo Gnoli. *Zoroaster’s Time and Homeland: A Study on the Origins of Mazdeism and Related Problems*. Number 7 in Series Minor. Istituto Universitario Orientale, Naples, 1980.
- John C. Gower. Generalized Procrustes analysis. *Psychometrika*, 40(1):33–51, 1975. doi: 10.1007/BF02291478.
- Richard S. Hess. *Studies in the Personal Names of Genesis 1–11*, volume 234 of *Alter Orient und Altes Testament*. Butzon & Bercker; Neukirchener Verlag, Kevelaer and Neukirchen-Vluyn, 1993.
- Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795, 1995. doi: 10.1080/01621459.1995.10476572.
- Maurice G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938. doi: 10.1093/biomet/30.1-2.81.
- Roland G. Kent. *Old Persian: Grammar, Texts, Lexicon*. American Oriental Society, New Haven, second edition, 1953.
- Ajinkya Kulkarni. Replication materials for “Geographic encoding in premodern enumerations: a five-corpus quantitative test”. <https://github.com/ajinkyakulkarni/geographic-encoding-enumerations>, 2026.
- Bimala Churn Law. *Tribes in Ancient India*. Bhandarkar Oriental Research Institute, Poona, second edition, 1973.

- Arthur A. Macdonell and Arthur B. Keith. *Vedic Index of Names and Subjects*, volume 2. John Murray, London, 1912.
- C. L. Mallows. Non-null ranking models. I. *Biometrika*, 44(1/2):114–130, 1957. doi: 10.1093/biomet/44.1-2.114.
- Nathan Mantel. The detection of disease clustering and a generalized regression approach. *Cancer Research*, 27(2):209–220, 1967.
- Roger Newson. Parameters behind “nonparametric” statistics: Kendall’s tau, Somers’ D and median differences. *Stata Journal*, 2(1):45–64, 2002. doi: 10.1177/1536867X0200200103.
- Pedro R. Peres-Neto and Donald A. Jackson. How well do multivariate data sets match? The advantages of a Procrustean superimposition approach over the Mantel test. *Oecologia*, 129(2):169–178, 2001. doi: 10.1007/s004420100720.
- T. W. Rhys Davids. *Buddhist India*. T. Fisher Unwin, London, 1903.
- Rudolf Roth. *Zur Litteratur und Geschichte des Weda*. A. Liesching & Comp., Stuttgart, 1846.
- Rüdiger Schmitt. *Wörterbuch der altpersischen Königsinschriften*. Reichert Verlag, Wiesbaden, 2014.
- Ajit Singh, Kristina J. Thomsen, Rajiv Sinha, Jan-Pieter Buylaert, Andrew Carter, Darren F. Mark, Philippa J. Mason, Alexander L. Densmore, Andrew S. Murray, Mayank Jain, Debajyoti Paul, and Sanjeev Gupta. Counter-intuitive influence of Himalayan river morphodynamics on Indus Civilisation urban settlements. *Nature Communications*, 8:1617, 2017. doi: 10.1038/s41467-017-01643-9.
- Barend A. van Nooten and Gary B. Holland. *Rig Veda: A Metrically Restored Text with an Introduction and Notes*, volume 50 of *Harvard Oriental Series*. Harvard University Press, Cambridge, MA, 1994.
- Gordon J. Wenham. *Genesis 1–15*. Word Biblical Commentary. Word Books, Waco, TX, 1987.
- Claus Westermann. *Genesis 1–11: A Commentary*. Augsburg, Minneapolis, 1984. Translated by John J. Scullion.
- Michael Witzel. R̥gvedic history: Poets, chieftains and politics. In George Erdosy, editor, *The Indo-Aryans of Ancient South Asia: Language, Material Culture and Ethnicity*, volume 1 of *Indian Philology and South Asian Studies*, pages 307–352. Walter de Gruyter, Berlin and New York, 1995.
- Michael Witzel. Substrate languages in Old Indo-Aryan (R̥gvedic, Middle and Late Vedic). *Electronic Journal of Vedic Studies*, 5(1):1–67, 1999.
- Michael Witzel. The home of the Aryans. In Almut Hintze and Eva Tichy, editors, *Anūsantatyai: Festschrift für Johanna Narten zum 70. Geburtstag*, number 19 in *Münchener Studien zur Sprachwissenschaft, Beiheft*, pages 283–338. J. H. Röhl Verlag, Dettelbach, 2000.
- Dai Yamazaki, Daiki Ikeshima, Jeison Sosa, Paul D. Bates, George H. Allen, and Tamlin M. Pavelsky. MERIT Hydro: A high-resolution global hydrography map based on latest topography dataset. *Water Resources Research*, 55(6):5053–5073, 2019. doi: 10.1029/2019WR024873.
- Yashpal, Baldev Sahai, R. K. Sood, and D. P. Agrawal. Remote sensing of the “lost” Sarasvati river. *Proceedings of the Indian Academy of Sciences (Earth and Planetary Sciences)*, 89(3):317–331, 1980.
- Heinrich Zimmer. *Altindisches Leben*. Weidmann, Berlin, 1879.